

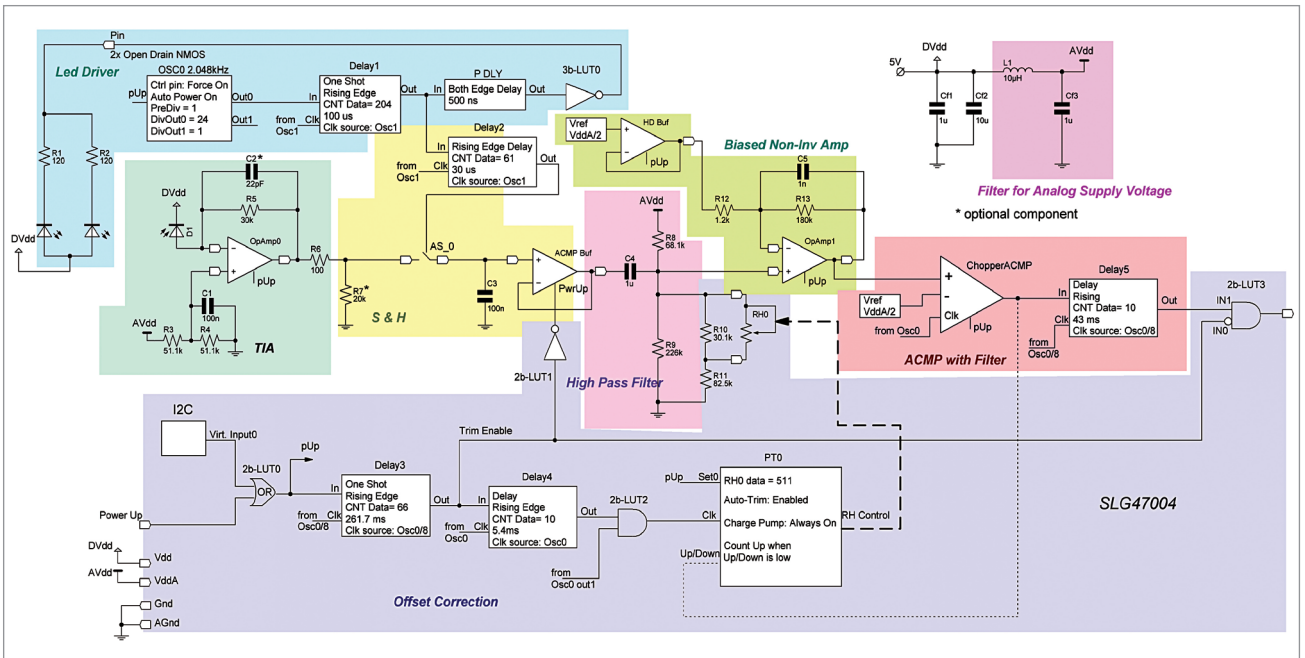


Fig. 4: The proposed analogue front-end structure (Credit: Dialog Semiconductor, a Renesas company)

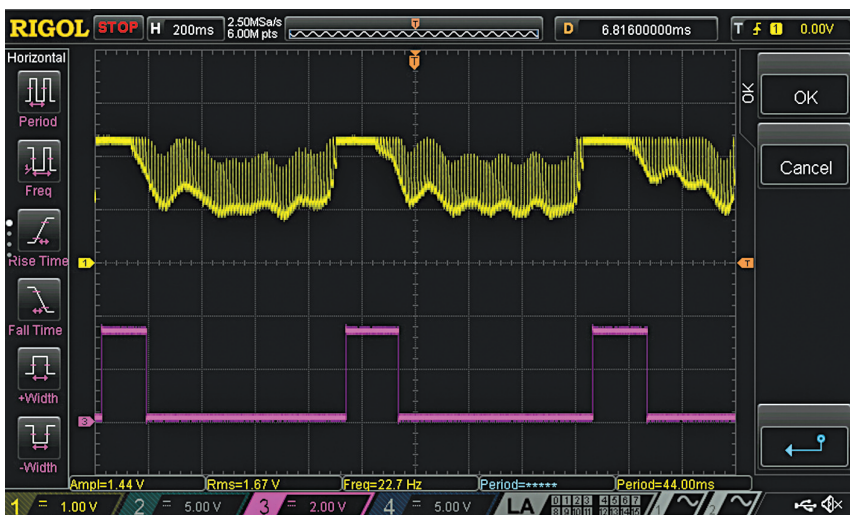


Fig. 5: Analogue (yellow) and digital (pink) output signals of AFE applied to finger (Credit: Dialog Semiconductor, a Renesas company)

means that it employs the principles of photoplethysmography. It can communicate with microcontrollers or any processing unit using an SPI interface.

AFE4403 contains an ADC, an LED transmit section, and diagnostics for sensor and LED fault detection. Moreover, it is equipped with an oscillator that functions from an external crystal. It is highly configurable and allows the user to

control the timing characteristics of the device.

AD4403 consists of five main blocks:

- Receiver channel
- Clocking and timing signal generation block
- Timer module
- Receiver subsystem power path
- Transmit section

The receiver channel receives the light signal from the photodiode

and contains a TIA, filters, and an ADC. The transmit section contains an LED driver that turns the IR and red LEDs on and off. What's interesting is that the transmit section has an option for a third LED to be attached, apart from the IR and red LED. This is particularly important for oximeter applications where the third LED can be used for heart rate monitoring while the other two are used for detecting SPO₂ levels.

SLG47004 AnaloguePAK by Dialog Semiconductor, a Renesas company. Dialog Semiconductor proposed a design of an AFE for a heart rate monitor that is largely based on SLG47004, a programmable mixed-signal IC. Both signal processing and LED control are performed by a single SLG47004. Additional sample and hold circuitry allow greatly reduced current consumption.

The performance of the heart rate monitor depends on the optical system setup: the direction of light from LEDs, the wavelength of LEDs, the distance and angle between LEDs and photodiode. Also, the measurements are affected by ambi-